

Quantum Mechanics III

Lecture 10

Continuous Measurement and Quantum Trajectories

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$d\rho = \mathcal{L}(\rho)dt + \sqrt{\eta}\mathcal{H}[c](\rho)dW$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$dY_t = \sqrt{\eta} \langle c + c^\dagger \rangle dt + dW$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$E[\rho_c(t)] = \rho(t)$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Park, ch. 10; Wiseman selections 4–5; QDC continuous-readout block